

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EC)

Apr-May 2019

EC245/EC207.01/EC207 Control Systems

Date: 06.05.2019, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1(a) Define Relative stability, Absolute stability and Conditional stability. [03]

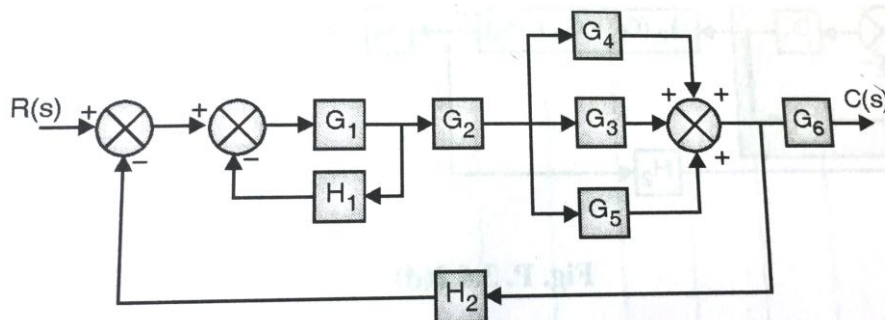
(b) Using Routh's criteria, for the unity feedback system, [04]

$$G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$$

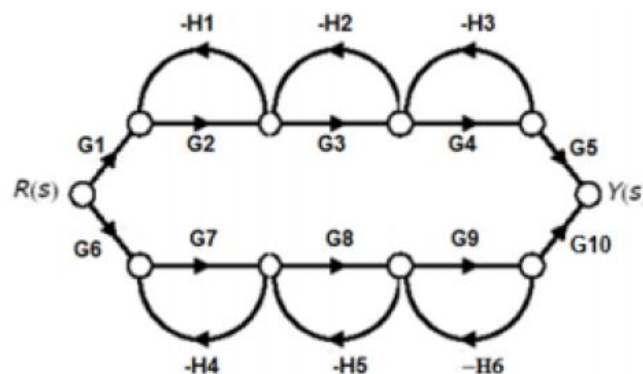
Find range of values of K.

Q – 2(a) Answer any two questions. [10]

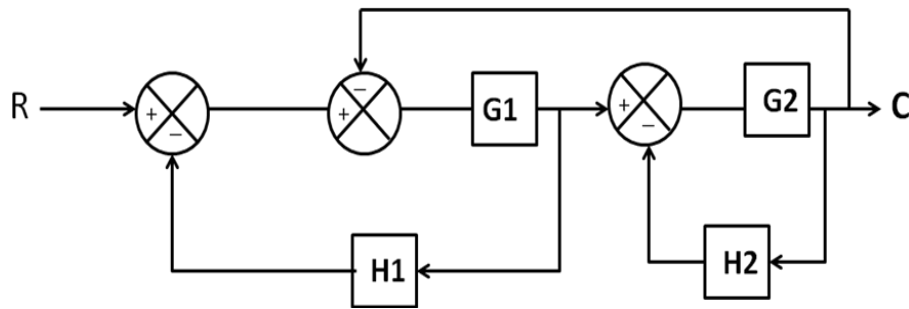
1) For the block diagram shown in figure below, determine the transfer function.



2) Obtain transfer function from the signal flow graph shown in following figure (Use Mason's gain formula).



- 3) Using block diagram reduction technique, find the closed loop transfer function of a system given below.



- (b) What are the different types of Controllers? Explain individual in details. [04]

Q – 3(a) Answer any two questions. [14]

- 1) A feedback Control System has an open loop transfer function

$$G(s)H(s) = \frac{K(s+4)(s+5)}{(s+1)(s-1)}$$

Sketch the Root Locus as K is varied from 0 to ∞ .

- 2) A feedback control system has characteristics equation

$$Q(s) = s(s+4)(s+8) + K(s+1)$$

Sketch the root locus for K as a variable.

- 3) A unity feedback control system has an open –loop transfer function of

$$G(s) = \frac{K}{s(s+1)(s+3)(s+4)}$$

Plot the Root Locus.

SECTION – II

Q - 4 (a) Do as directed. [04]

- 1) Derivation of Peak Time
- 2) Derivation of Peak Overshoot

- (b) Define the following terms. [03]

- 1) Transient Response
- 2) Steady State Error
- 3) Rise Time

Q - 5 (a) Sketch the Bode plot for the transfer function. [05]

$$G(s)H(s) = \frac{s^2}{(1+0.2s)(1+0.02s)}$$

OR

- (a) Sketch the bode plot and determine the gain crossover and phase cross over frequencies for the transfer function. [05]

$$G(s)H(s) = \frac{10}{s(1+0.5s)(1+0.1s)}$$

- Q - 5 (b) 1) A unity feedback system has [05]

$$G(s) = \frac{40(s+2)}{s(s+1)(s+4)}$$

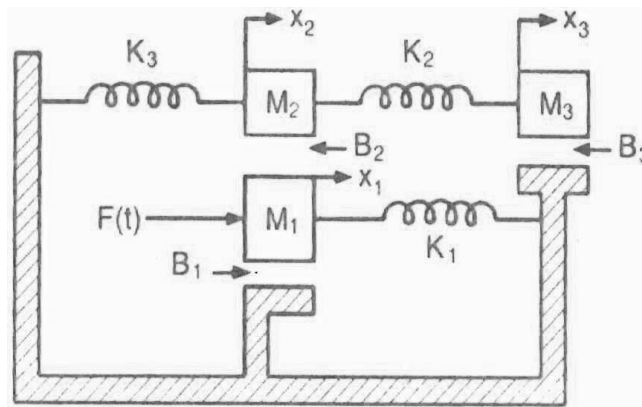
Determine

- Type of system
- All error coefficients
- Error for ramp input with magnitude 4.

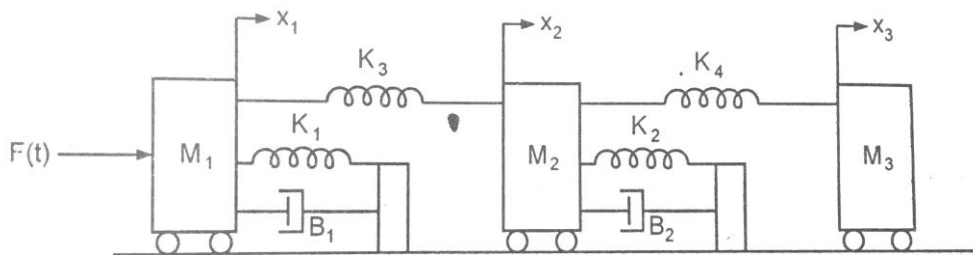
- Q - 5 (c) For a system $G(s)H(s) = \frac{K}{s^2(s+2)(s+3)}$, find the value of K to limit steady state error [04]

to 10 when input to system is $1+10t+\frac{40}{2}t^2$.

- Q - 6 (a) For the following translation system shown in figure, obtain the differential equations. [03]



- Q - 6 (b) Draw the mechanical network for the mechanical system shown in figure below. [06]
Obtain an analogues electrical circuit based on Force- Current analogy.



Q - 6 (c) Find the stability of the following system using Nyquist Criteria.

[05]

$$G(s) = \frac{60}{(s+1)(s+2)(s+5)}$$

OR

(c) Using Nyquist criterion, examine the closed loop stability of a system whose open loop transfer function is **[05]**

$$G(s) = \frac{10(s+3)}{s^2 - 4}$$
